

Data Card — AgriAdviser-NG

A domain-specific small language model (SLM) for climate-aware rice & maize advisory to smallholder farmers in Nigeria

Input vs. Output

Inputs: Free-text questions typed or sent by farmers over USSD/SMS (primary channel) or WhatsApp (secondary channel), in Nigerian English and Nigerian Pidgin. Inputs are short, often informal, and may describe a symptom rather than ask a direct question (e.g., “my rice leaf dey turn yellow for the edge”).

Outputs: A short, actionable text advisory response grounded in retrieved source material, a recommendation, a diagnosis with next steps, or an explicit abstention (“I’m not sure, please contact your local extension officer”) when no confident match is found in the retrieval layer.

Classification problem

AgriAdviser-NG is not primarily a classification system, it's a retrieval-augmented generation task: match a farmer's question to the most relevant source passage(s), then generate a grounded answer from that context.

That said, a lightweight classification step sits underneath the retrieval layer: incoming questions are implicitly routed by topic/category (e.g., pest identification, soil/fertiliser guidance, water management, climate/planting-window advice, post-harvest handling) and by crop (rice vs. maize), since these are the fields the retrieval corpus is organized around. Getting this routing right matters because a wrong topic/crop match is the most common way an otherwise-fluent answer becomes actively wrong.

Labeling

Labels here mean two things: (1) the source-document metadata (topic, crop, agro-ecological zone) attached to each reference document, and (2) the reference answers used to fine-tune and evaluate the model.

Source: A mix of domain-expert material (state ADP fertilizer tables, IITA/AfricaRice variety bulletins, NAERLS extension manuals, NiMet rainfall bulletins) and, where real farmer-question data is scarce, carefully scoped synthetic question generation grounded in that same expert material, never invented facts, only invented phrasings of real, sourced content.

Consistency/accuracy checks: Every reference answer must be traceable to a specific source passage before it's added to the training set — a “no source, no answer” rule. Synthetic and real/extracted rows are explicitly tagged (an origin field distinguishing synthetic vs. extracted, plus a source_url/citation), so quality issues can be traced back to a specific source and provenance is never ambiguous later. Where possible, a second reviewer (ideally someone with agronomic domain knowledge, not just the developer) should spot-check a sample of labels against the source material before each training run.

Data creation

Direct sourcing: Extracting and paraphrasing real content from official extension bulletins, variety guides, and climate advisories (IITA, AfricaRice, NAERLS, NiMet, state ADPs). This is the trustworthy backbone of the dataset, real agronomic facts, properly attributed.

Synthesis for low-resource gaps: Nigerian Pidgin and informal Nigerian-English phrasings of farmer questions are under-represented in formal extension literature, which is written in standard English for extension agents, not farmers. Where authentic farmer-voice question data isn't available, synthetic paraphrases are generated from the same verified source content, varying phrasing, formality, and dialect, rather than inventing new facts. This directly serves the goal of the model actually understanding how farmers speak, not just how agronomists write.

A future, higher-priority addition: authentic farmer questions collected through the extension-officer/community channel itself (see Bias and Representation below), rather than relying on synthetic Pidgin approximations indefinitely.

Web collection ethics

Sourcing is limited to public research and extension-institute repositories: IITA, AfricaRice, NAERLS, and NiMet, plus state-level ADP publications. No general web scraping, no social media, no farmer forums are used.

Consent: These are institutional publications intended for public agricultural extension use, not personal data, no individual consent question arises from the documents themselves.

Copyright: Most of this material is not openly licensed (typically “all rights reserved, reproduce with permission” rather than public domain or CC0). Before any production or public deployment, each source's actual reuse terms need to be checked and, where required, permission requested, a paraphrased, attributed summary used for research/prototyping is a different bar than redistributing content in a shipped product.

Privacy: Once real farmer-question data is collected (via extension officers or a pilot deployment), farmer identity, phone number, and location must be stripped or anonymized before any question is added to a training set, this is a hard requirement, not a nice-to-have, given the sensitivity of linking someone's specific farm/location to their crop-failure questions.

Bias and representation

Several groups are at real risk of being underserved by this dataset as currently scoped:

- **Dialect/language coverage:** Formal extension English dominates the source corpus; Pidgin and regional dialect phrasings are currently synthetic approximations, not authentic farmer speech, this is the dataset's biggest current representation gap.
- **Geographic/agro-ecological skew:** Much of the readily available IITA/AfricaRice/NAERLS material leans toward specific agro-ecological zones (often semi-arid/Sahelian contexts for widely studied crops like sorghum and groundnut); rice- and maize-specific coverage across all of Nigeria's relevant zones (sub-humid, highland, etc.) needs deliberate checking for gaps, not assumed.
- **Literacy and access:** The USSD/SMS/text-first design already excludes farmers who can't read, and the explicit exclusion of a voice interface (per project scope) compounds this — a real tension between the project's offline-first, low-bandwidth design goal and full inclusivity that's worth stating openly rather than glossing over.
- **Gender:** Agricultural extension data collection has a well-documented historical tendency to under-represent women farmers' questions and priorities; without deliberate attention, a dataset built from existing extension literature risks inheriting that same skew.

Mitigation approach: the project's built-in abstention behavior (explicitly saying “I don't know, contact an extension officer” rather than guessing) is itself a partial bias mitigation, it's safer for the system to admit low confidence for an under-represented dialect/context than to confidently generate a plausible-sounding but ungrounded answer.

Values and community alignment

This dataset design is built around the same values that shaped the original problem statement:

Augmentation, not replacement: the dataset and model are explicitly scoped to support, not replace, human extension officers, reflected in the abstention design and the decision to keep source material always traceable, so an extension officer (or the farmer) can always ask “based on what?”

Offline-first as a form of equity: designing for edge deployment (CPU-only, 2–3GB VRAM, GGUF quantized) instead of assuming smartphone/broadband access is a direct response to the reality of connectivity in the communities this is meant to serve, the dataset and model need to work without assuming a data plan.

Transparency over convenience: tagging every row's origin (synthetic vs. extracted) and source is more work than treating all rows the same, but it's what makes the dataset auditable, a community member, policymaker, or future developer should be able to ask “where did this specific piece of advice come from?” and get a real answer, not “the model said so.”

Accountability: as this moves from prototype toward real deployment, the intended accountability loop is: source material → traceable reference answer → model output → farmer, with an explicit off-ramp to a human extension officer at the point where the model's confidence (and the dataset's coverage) runs out — rather than a model that answers everything regardless of whether it actually knows.